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United States Patent	12406569
Kind Code	B1
Date of Patent	September 02, 2025
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Pull station housing

Abstract

The present invention comprises a pull station housing. A baseplate of the pull station housing may be attached to a surface. A frame may be removably attached to the baseplate and may form a perimeter around an alarm pull station. A housing magnet may be attached to the frame. A cover may be hingedly attached to the frame and may be separated from the baseplate by at least the frame. The cover may cover at least one side of the alarm pull station. A cover magnet of opposite polarity than the housing magnet may be connected to the cover and may be removably attached to the housing magnet by magnetic attraction. The alarm pull station may be accessed by rotating the cover away from the frame, thereby breaking the magnetic connection between the cover magnet and the housing magnet.

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Family ID:	96882253
Appl. No.:	19/185076
Filed:	April 21, 2025

Publication Classification

Int. Cl.: G08B25/12 (20060101); G08B17/06 (20060101)

U.S. Cl.:

CPC G08B25/12 (20130101); G08B17/06 (20130101);

Field of Classification Search

CPC: G08B (25/12); G08B (17/06)

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention relates to housings for alarm pull stations. Alarm pull stations are devices typically mounted to a wall that, when activated, cause an alarm to emit a noise. Alarm pull stations are typically electrically connected to an alarm system with multiple alarms spaced about a structure such as a school, office building, etc. Alarm pull stations are commonly used as fire alarms. If a user suspects a fire in a building, the user may activate the alarm pull station to cause one or more of the alarms to emit a noise, thereby warning others in the building of the fire.

(2) Alarm pull stations are typically activated by pulling a lever. Pulling the lever causes an electrical connection to be sent to one or more alarms of the alarm system to which the alarm pull station is connected. Due to the electrical components required for the alarm pull station to function, alarm pull stations are often secured by a protective housing. The protective housing further serves to prevent the alarm pull station from being accidentally activated.

(3) Most housings for alarm pull stations have a cover that is either broken by a user or moved out of the way to access the alarm pull station. In housings wherein the cover is broken to access the alarm pull station, there is a risk of broken glass or plastic that may harm the user that activates the alarm pull station. In housings wherein the cover is moved out of the way to access the alarm pull station, the cover is often secured in place with a snap feature that is difficult to engage to close the cover and is easily broken while opening the cover.

(4) Many housings for alarm pull stations emit a noise and/or light when their cover is lifted to warn a user that the cover has been removed from the alarm pull station. The noise emitted by the alarm pull station housing is often quieter than the noise emitted by the alarms connected to the alarm pull station. In housings that emit light when their cover is lifted, multiple lights are often positioned around the housing however there are none that lights from the sides of the housing.

(5) Traditional pull station covers often rely on snap-on mechanisms that can fail to secure properly, leading to accidental activations and increased maintenance costs. Additionally, existing designs frequently use screws to attach components, which creates potential entry points for moisture, reducing the effectiveness of the seal and increasing the risk of environmental damage. The current lack of side lighting in fire alarm covers can also hinder visibility at certain angles of view. Furthermore, servicing these covers is often cumbersome, requiring manual removal of security pins or screws.

SUMMARY OF THE INVENTION

(6) The present invention addresses the above and other needs by providing a fire alarm pull station

housing comprising a baseplate, a frame, a cover, a cover handle, a magnetic engagement means, and an alarm assembly. There are two types of mounting configuration for the pull station housing. The first is a flush mount configuration and the second is a surface mount configuration. In the flush mount configuration, the baseplate is also the frame. The baseplate is removably attached to a wall and forms a perimeter around the fire alarm pull station; the cover is hingedly attached to the baseplate; the magnetic engagement means engages the cover and the baseplate such that the cover is securely in the closed position. When a pull force is exerted upon the handle, sufficient to overcome the resistance of the magnetic engagement means, the cover is displaced laterally away from the frame in an open position triggering the alarm assembly. The magnetic engagement means secures the cover in the closed position after the alarm assembly has been reset.

(7) In the surface mount configuration, the baseplate is attached to the wall; the frame, which forms a perimeter around the fire alarm pull station, is attached to the baseplate; and the cover is hingedly attached to the frame by the magnetic engagement means.

(8) In various embodiments, the fire alarm pull station housing includes a magnetic engagement means for securing the cover in a closed position relative to the frame and/or baseplate. The magnetic engagement means may be configured to resist displacement of the cover under normal conditions and to release the cover when a predetermined force is applied, such as a pulling force on a handle or direct manual contact with the cover.

(9) The magnetic engagement means may include permanent magnets, electromagnets, magnetic latches, or magnetic couplings. The magnets may be positioned on the interior surface of the cover and aligned with complementary magnetic or magnetically-attractable elements (e.g., ferromagnetic plates or steel inserts) located on the frame, baseplate, or other pull station housing component.

(10) In one embodiment, one or more magnets are embedded or affixed within the cover, and corresponding ferrous components are integrated into the frame, thereby forming a releasable magnetic bond. Alternatively, magnets may be placed on the frame, with the cover comprising a ferrous or magnetically reactive engagement plate.

(11) In some embodiments, the magnetic engagement means includes multiple magnetic contact points to increase holding strength or ensure uniform sealing around the perimeter of the cover. The magnets may be arranged symmetrically or asymmetrically depending on the desired force profile and accessibility requirements.

(12) In other embodiments, the magnetic engagement means includes a selectively activatable magnetic release mechanism, wherein an electromagnet can be energized to release or secure the cover under electronic control.

(13) The magnetic engagement means may further incorporate adjustable or tunable features, allowing the strength of the magnetic holding force to be adapted to specific applications or user preferences. For example, a shim or spacer element may be used to modulate the effective gap between magnetic elements, thereby adjusting the release force threshold.

(14) In all embodiments, the magnetic engagement means is configured such that when a user exerts a force on the cover, such as pulling a handle, the magnetic retention is overcome and the cover moves to an open position, triggering an associated alarm assembly. After the alarm is deactivated or reset, the magnetic engagement means allows the cover to be repositioned and secured in the closed state without the need for mechanical latches or fasteners.

(15) The magnetic engagement means may be used alone or in combination with mechanical hinges, biasing elements (e.g., springs), or dampers to provide smooth opening/closing action and controlled resistance.

(16) The magnetic engagement means having a magnetic strength to compress a sealing member that is interposed between the cover and the frame, and wherein the magnetic engagement means urges the cover toward the frame, compressing the sealing member, thereby forming a substantially airtight and dust-resistant seal.

(17) In accordance with one embodiment of the invention, there is provided a pull station housing with a cover hingedly attached to a frame or to a baseplate. The cover may have 5 cover sides, each with a cover side length and a cover side width. Certain cover sides may have different cover side lengths and cover side widths than other cover sides; i.e. some of the cover sides may be larger than others. Each of the 5 cover sides may be flat. The 5 cover sides may be joined by cover edges. The cover edges may be rounded and may each have a radius of curvature. The radii of curvature of the cover edges may be less than the smallest cover side width.

(18) In accordance with one embodiment of the invention, the cover may be hingedly attached to the frame. In accordance with another embodiment of the invention, the pull station housing may not have a frame, and the cover may be hingedly attached to the baseplate. In all embodiments, the cover may cover at least one side of an alarm pull station. In embodiments of the pull station housing without the frame, the cover may form the perimeter around the alarm pull station that the frame forms in embodiments of the pull station housing with the frame.

(19) In accordance with another embodiment of the invention, the alarm assembly having a housing; a sound emitter and a light source integrated within the alarm assembly housing. The sound emitter is configured to emit a noise when the cover is in the open position. The light source is also configured to emit light along the right and left sides of the alarm assembly housing when the cover is in the open position.

(20) In accordance with another embodiment of the invention, the cover has an exterior surface and an interior surface. The interior surface of the cover having a plurality of nubs. The nubs are configured such that the alarm assembly is tightly held to the interior cover surface by the nubs by interference fit.

(21) In accordance with another embodiment of the invention, an alarm delay mechanism is configured on the alarm assembly and may be electrically coupled to the sound emitter and/or the light source. When the cover of the pull station housing is in the open position, the sound emitter of the alarm assembly emits a loud noise. Pressing the delay assembly causes the sound emitter to cease emitting noise for a preprogrammed time period. This may be accomplished by stopping the power supply to the noise emitter or by activating a switch that causes a current to stop flowing to the sound emitter. Pressing the delay mechanism will also cause the light to cease emitting light for the same period as the noise emitter.

(22) In accordance with another embodiment of the invention, a frame sealing ring may form a seal between the frame and the baseplate; a cover sealing ring may form a seal between the cover and the frame; the cover sealing ring may form a seal between the cover and the baseplate. The cover sealing ring and the frame sealing ring may be an elastic material such as but not limited to silicone, rubber, or the like.

(23) In accordance with another embodiment of the invention, a conduit opening is configured in the frame, through the baseplate, and between the frame and the baseplate. A conduit insert surrounds the conduit opening such that the conduit insert is configured in an opening within the frame, within the baseplate, or between the frame and the baseplate. A conduit is inserted through the conduit opening. The conduit wires connect to the alarm pull station and/or other electrical components of the pull station housing. The conduit insert holds the conduit in place. In some embodiments, the conduit insert may form a seal around the conduit. In some embodiments, the conduit insert may form a seal between the baseplate and the frame.

(24) In accordance with another embodiment of the invention, a subplate is configured within the frame. The subplate separates the alarm pull station from the baseplate. The subplate having a subplate cutout that reveals at least a portion of the alarm pull station to the baseplate. The subplate cutout allows for access to the alarm pull station if electrical wires or other components are connected to the alarm pull station. The subplate cutout further allow access to the alarm pull station if temperature components such as heaters or fans are configured within the frame between the subplate and baseplate. In accordance with some embodiments, the subplate cutout are the same

shape as the subplate.

(25) In accordance with another embodiment of the invention, a fan is configured between the subplate and the baseplate. The fan removes moisture from the pull station housing by blowing moist air out of the pull station housing and/or by blowing dry air into the pull station housing. The fan regulates the temperature within the pull station housing by blowing cold or hot air onto the alarm pull station.

(26) In accordance with another embodiment of the invention, a heater is configured between the subplate and the baseplate. The heater removes moisture from the pull station housing by heating the air within the pull station housing, thereby causing any water inside of the pull station housing to evaporate and be pushed out of the pull station housing by the fan. The heater regulates the temperature within the pull station housing by providing a source of heat when the pull station housing is implemented in a cold area such as an exterior wall of a building.

(27) Both the fan and the heater may be considered “temperature components” of the pull station housing. These temperature components aid in regulating the temperature and humidity within the pull station housing. This is beneficial in ensuring that the alarm pull station, the noise emitter, the light source, and any other electrical components contained within the pull station housing function as intended throughout their intended lifespan. The fan and the heater may work in tandem to regulate the humidity within the pull station housing by evaporating water and then blowing the evaporated water out of the pull station housing. The fan and the heater work in tandem to regulate the temperature within the pull station housing by heating some of the air within the pull station housing and then blowing the heated air throughout the pull station housing, thereby heating the entire pull station housing.

(28) In accordance with another embodiment of the invention, a plurality of subplate offsetters separate the subplate and the baseplate. The plurality of subplate offsetters may be 2 or more subplate offsetters, 4 or more subplate offsetters, 6 or more subplate offsetters, or any number between or including the values provided. Threaded fasteners thread through the subplate, through the subplate offsetters, and into the baseplate or frame to secure the subplate to the rest of the pull station housing. The subplate offsetters separate the subplate from the baseplate by $\frac{1}{2}$ inch or more, 1 inch or more, 2 inches or more, or any distance between or including the values provided. The subplate offsetters allow for the fan and/or heater to be configured between the baseplate and the subplate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features and advantages of the present invention will be more apparent from the following more particular description thereof, presented in conjunction with the following drawings wherein:

(2) FIG. 1 is a perspective exploded view of a pull station housing in a surface mount configuration according to the present invention.

(3) FIG. 2 is a perspective exploded view of a pull station housing in the flush mount configuration according to the present invention.

(4) FIG. 3 is a front view of a pull station housing according to the present invention.

(5) FIG. 4 is a right side view of a pull station housing in a closed position according to the present invention.

(6) FIG. 5 is a right side view of a pull station housing with a cover configured in an open position according to the present invention.

(7) FIG. 6 is a top view of a pull station housing according to the present invention.

(8) FIG. 7 is a right side cross-section view of a pull station housing along lines 7-7 of FIG. 6,

according to the present invention.

(9) FIG. **8** is a rear view of a pull station housing according to the present invention.

(10) FIG. **9A** is a top view of a conduit insert according to the present invention.

(11) FIG. **9B** is a side view of the conduit insert according to the present invention.

(12) FIG. **9C** is a top view of a solid conduit insert according to the present invention.

(13) FIG. **10** is a top view of the pull cover sliding into the groove of the frame according to the present invention.

(14) FIG. **11** is a perspective view of the back of the cover according to the present invention.

(15) FIG. **12** is a rear view of the cover according to the present invention.

(16) FIG. **13** is a horizontal exploded view of the cover and the cover sealing ring according to the present invention.

(17) FIG. **14** is a perspective view of the surface mount frame according to the present invention.

(18) FIG. **15** is a perspective view of the flush mount frame according to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(19) The following description is of the best mode presently contemplated for carrying out the invention. This description is not to be taken in a limiting sense, but is made merely for the purpose of describing one or more preferred embodiments of the invention. The scope of the invention should be determined with reference to the claims.

(20) Where the terms “about” or “generally” are associated with an element of the invention, it is intended to descript a feature's appearance to the human eye or human perception, and not a precise measurement, or typically within 10 percent of a stated value.

(21) As shown in FIG. **1**, the fire alarm pull station housing **10** comprising a frame **30** of a pull station housing **10** is attached to a baseplate **20** of the pull station housing **10**. A frame sealing ring **33** is configured between the frame **30** and the baseplate **20** to form an airtight and watertight seal between the frame **30** and the baseplate **20**. Conduit openings **22** are configured through the baseplate **20**. The conduit openings **22** may allow a conduit that surrounds one or more wires to be passed through the baseplate **20** so that the wires may be coupled to various electrical components of the pull station housing **10**. A temperature component **64** is configured within the frame **30**. The temperature component **64** may be any device capable of regulating temperature and/or humidity levels within the pull station housing **10**. The temperature component **64** may be a fan, a heater, or the like. The temperature component **64** may be electrically coupled to the wires of the conduit.

(22) A subplate **60** is attached to the baseplate **20**. Subplate offsetters **68** separate the baseplate **20** and the subplate **60** so that components such as the temperature component **64** is configured between the baseplate **20** and subplate **60**. Threaded fasteners are threaded through the subplate **20**, through the subplate offsetters **68**, and into the baseplate **20** to attach the subplate **60** to the baseplate **20**. In some embodiments, the threaded fasteners are threaded through the subplate **60**, through the subplate offsetters **68**, and into the frame **30** to attach the subplate **60** to the frame **30**. In these embodiments the threaded fasteners may thread into a back portion of the frame that is adjacent to the baseplate **20**.

(23) An alarm pull station **14** is attached to one face of the subplate **60** such that the subplate **60** separates the alarm pull station **14** from the baseplate **20**. The subplate **60** is situated within the frame **30** such that the alarm pull station **14** is completely situated within the frame **30**.

Alternatively, the subplate **60** may be situated within the frame **30** such that only a portion of the alarm pull station **14** is situated within the frame **30**. The depth at which the subplate **60** is situated in the frame **30** is determined by the length of the subplate offsetters **68**. The subplate **60** has a subplate cutout **62** that extends through the subplate **60**. This allows at least a portion of the alarm pull station **14** to be exposed to the baseplate **20**. This allows wires to be connected to the alarm pull station **14**. This also allows the temperature component **64** to effectively regulate the temperature and/or humidity experienced by the alarm pull station **14**.

(24) The alarm pull station **14** may be any device that, when activated, causes an alarm to be

activated. Most pull stations in the art are electrically connected to alarms. A lever on the alarm pull station **14** may be pulled to cause an electrical signal to be sent to the alarms, causing the alarms to emit a noise. The electrical and mechanical components of the alarm pull station **14** may be sensitive and may need to be protected. Therefore, the pull station housing **10** provides a physical barrier between the alarm pull station **14** and an exterior environment.

(25) A cover **40** of the pull station housing **10** is hingedly attached to the frame **30**. A cover sealing ring **43** is configured between the cover **40** and the frame **30** to form an airtight and watertight seal between the cover **40** and the frame **30**. The cover **40** may rotate relative to the frame **30** at a hinge **70** (See FIGS. **4** and **5**) to shield the alarm pull station **14** or to expose the alarm pull station **14**. The cover **40** may be transparent so that the alarm pull station **14** is visible through the cover **40** (See FIG. **3**). The cover **40** may be made of glass, plastic, or any other transparent material. A lift tab **47** may extend from the cover **40** and may be grasped when rotating the cover **40** relative to the frame **30**.

(26) A magnetic engagement means comprising a cover magnet **42** attached to an inside surface of the cover **40** and a housing magnet **32** attached to the frame **30**. Either the cover magnet **42** or the housing magnet **32** made be a metal piece. When a pull force is exerted upon the handle sufficient to overcome the resistance of the magnetic engagement means, the cover is displaced laterally away from the frame in an open position triggering the alarm assembly. Moreover, when the alarm assembly is reset, the magnetic engagement means secures the cover to be in the closed position.

(27) An alarm assembly **50** is configured between the cover **40** and the subplate **60**. The alarm assembly **50** houses certain electrical components of the pull station housing **10**, including but not limited to a sound emitter **54** and a light source **56**. When the cover **40** is opened by rotating the cover **40** away from the frame **30** and thereby breaking the magnetic connection between the cover magnet **42** and the housing magnet **32**, the light **56** may flash and the sound emitter **54** may emit a noise to warn that the cover **40** has been opened.

(28) As shown in FIG. **2**, in a flush mount configuration, the cover **40** is hingedly attached to the baseplate **20**. The baseplate **20** and base plate seal ring are attached to the wall **16**. The housing magnet **32** is attached to the baseplate **20** instead of the frame **20**. The cover **40** is opened by rotating the cover **40** away from the baseplate **20**. This may still cause the light source **56** of the alarm assembly **50** to flash and the sound emitter **54** of the alarm assembly **50** to emit a noise. The alarm pull station **14** is still contained within the pull station housing **10**. As shown, the alarm pull station **14** is still configured between a subplate **60** and the cover **40**. The subplate **60** is still shown with a subplate cutout **62** so that electrical wires can be coupled to the alarm pull station **14** through the subplate **60**. When the cover **40** is closed, the cover sealing ring **43** may form an airtight and watertight seal between the cover **40** and the baseplate **20**.

(29) As shown in FIG. **3**, the baseplate **20** of the pull station housing **10** is attached to a surface **16**. The attachment of the baseplate **20** to the surface **16** may be accomplished by nails, threaded fasteners, hook-and-loop fasteners, adhesive, or any other means for attaching a component to a surface. In most embodiments, the surface **16** is a wall, though this is not intended to limit the surface **16** to being a wall in all embodiments. The surface **16** may be a fence, floor, ceiling, door, window, or other feature on which it may be beneficial to install an alarm pull station and corresponding housing. In embodiments wherein the surface **16** is a wall, the surface **16** may be an exterior wall or an interior wall.

(30) The cover **40** has five cover sides **44**, each connected to at least one other cover side by a cover edge **48**. The cover sides **44** may be flat or may have one or more curvatures. The cover edges **48** are preferably rounded.

(31) The hinge **70** having a hinge pin **72**, a cover hinge component **74** and a frame hinge component **76**. The hinge **70** is shown that hingedly connects the cover **40** to the baseplate **20** or frame (not visible in FIG. **3**) and the cover **40** rotates about the baseplate **20** along the hinge **70**.

(32) The alarm pull station **14** and alarm assembly **50** are visible through the cover **40** due to the

cover **40** being transparent. Text is provided on the alarm pull station **14** and alarm assembly **50** to aid a user in using the alarm pull station **14** to activate an alarm. Text is also provided on the lift tab **47** to aid a user in lifting the cover **40** to expose the alarm pull station **14**.

(33) A delay mechanism **55** is provided on the alarm assembly **50**. The delay mechanism **55** protrudes from the alarm assembly **50** so that it may be accessed without opening the alarm assembly **50**. When the cover **40** is lifted by rotating the cover **40** away from the baseplate **20** or frame, the alarm assembly **50** emits light and/or noise to alert a user that the cover **40** has been opened. This is beneficial if the alarm pull station **14** is being used to activate an alarm, or if the cover **40** has been lifted unintentionally. However, if maintenance is being performed on the alarm pull station **14**, then it would not be beneficial for light and/or noise to be emitted from the alarm assembly **50**. Therefore, the delay mechanism **55** may be engaged to deactivate the electrical components within the alarm assembly that emit light and/or noise. When maintenance is performed on the alarm pull station **14**, the cover **40** may be lifted to expose the alarm pull station **14** and the delay mechanism **55** may be engaged to cease the light and/or noise being emitted from the alarm assembly **50** due to the cover **40** being in the open position.

(34) As shown in FIGS. **4** and **6**, the cover having sides **44** and side edges **48**. The edges **48** having a lateral radius of curvature **R1** (See FIG. **6**) and a vertical radius of curvature **R2** (See FIG. **4**). **R1** may be from 0.500 to 1.000 inches, preferably about 0.875, most preferably 0.875 inches. **R2** may be from 0.15 to 0.50 inches, preferably about 0.25 inches, and most preferably 0.25 inches.

(35) As shown in FIGS. **4** and **5**, the alarm pull station **14** may extend from the frame **30** so that it is at least partially visible from a side view of the cover **40**. The alarm assembly **50** having sides **52** is visible from a side view of the cover **40**. In FIG. **4**, the cover **40** is shown as closed (i.e. in a closed position). The cover magnet **42** and housing magnet **32** are contacting each other to keep the cover **40** in the closed position. In FIG. **5**, the cover **40** is shown as opened (i.e. in an open position). The cover magnet **42** and housing magnet **32** are not contacting. The cover **40** is “lifted” such that it rotates about the hinge **70** away from the frame **30** and away from the baseplate **20**. In an open position, the alarm pull station **14** is accessible for use or for maintenance.

(36) The alarm assembly **50** having side openings **58** are configured in the alarm assembly right side and left side. When the cover **40** is opened and light is emitted from the alarm assembly **50**, light from the alarm assembly **50** is visible from a side view.

(37) As shown in FIG. **6**, the conduit opening **22** is configured through a conduit insert **24**. The conduit insert is configured in an opening within the frame **30**, but may also be configured between the frame **30** and the baseplate (not shown in FIG. **6**). The conduit insert **24** may be made of foam when the pull station housing **10** is attached to an interior surface of a structure. The conduit insert **24** may be made of plastic when the pull station housing **10** is attached to an exterior surface of a structure. The plastic material of the conduit insert **24** may provide a better barrier against debris, extreme temperatures, and water that may affect the functionality of the alarm pull station. The foam material of the conduit insert **24** may expand within its respective opening in the frame or between the frame or baseplate and therefore may form a better seal than the plastic material of the conduit insert. In some embodiments, the conduit insert may be made of both foam and plastic and may be used when the pull station housing **10** is attached to an interior surface or an exterior surface.

(38) As shown in FIG. **7**, the sound emitter **54** and the light **56** are housed within the alarm assembly **50**. The subplate **60** is connected to a rear face of the frame **30** and is separated from said rear face by the subplate offsetters **68**. The temperature component **64** is configured between the subplate **60** and the baseplate **20**. The alarm pull station **14** and temperature component **64** are exposed to one another via the subplate cutout **62**. The conduit opening **22** is configured in the frame **30** and allows a conduit to be inserted into the frame **30**.

(39) As shown in FIG. **8**, cover nubs **41** are configured on the interior of the cover **40**. The cover nubs **41** are configured in a pattern about the cover **40** to form a space into which the alarm

assembly **50** is placed. The alarm assembly **50** may be held against the cover **40** by an interference fit with the cover nubs **41**. This allows the alarm assembly **50** to be tightly held to the cover **40** without the need of a screw which compromises the airtight seal from moisture or debris. The movement of the alarm assembly **50** away from the frame **30** or baseplate **20** along the hinge **70** when the cover **40** is opened may trigger the light of the alarm assembly to emit light and/or the sound emitter of the alarm assembly **50** to emit noise. The nubs **41** are configured in such a way to provide enough space for the alarm pull station **14** to be covered by the cover **40** along with the alarm assembly **50**. The nubs **41** are configured in such a way as to not interfere with the placement of the alarm pull station **14** or to interfere with a user accessing the alarm pull station **14**.

(40) FIG. **9A** is a top view of the foam conduit insert **24** with a conduit opening **22** and a conduit flange **26** configured to fit tightly within the conduit groove **28** (See FIG. **10**); and FIG. **9C** is a top view of a solid foam conduit insert **24**. As shown in FIG. **10**, the foam conduit insert **24** engages the frame **30**, wherein the conduit insert flange **26** engages the conduit groove **28** such that it creates an airtight seal of the pull cover **10** when it is attached to a wall or a surface **16**.

(41) FIGS. **11** and **12** show the rear view of the cover **40** having a groove **36**, nubs **41**, and handle **47**.

(42) FIG. **13** is an exploded view of the cover **40** and the seal ring **43**, wherein the seal ring **43** tightly engages the groove **36** by an interference fit.

(43) FIG. **14** is a perspective view of the surface mount frame **30** having tongue **66** and FIG. **15** is a perspective view of the flush mount frame **30** having tongue **66**. When the pull station housing **10** is in the closed position (See FIG. **4**), the tongue **66** engages the groove **36**, wherein the tongue **66** compresses the seal ring **43** within the groove **36** by the magnetic engagement means.

(44) While the invention herein disclosed has been described by means of specific embodiments and applications thereof, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope of the invention set forth in the claims.

Claims

1. A fire alarm pull station housing comprising: a baseplate removably attached to a surface; a frame removably attachable to the baseplate and forming a perimeter around an alarm pull station; a cover hingedly attached to the frame; a handle secured to the cover; a magnetic engagement means; an alarm assembly, wherein when a pull force is exerted upon the handle sufficient to overcome the resistance of the magnetic engagement means, the cover is displaced laterally away from the frame in an open position triggering the alarm assembly; and the magnetic engagement means securing the cover to the frame when no pull force is exerted.
2. The pull station housing of claim 1, wherein a sealing member is interposed between the cover and the frame, and wherein the magnetic engagement means urges the cover toward the frame, compressing the sealing member, thereby forming a substantially airtight and dust-resistant seal.
3. The pull station housing of claim 1, wherein the cover further having: a cover front and a cover rear; the cover rear having a groove along the perimeter to receive the frame; and a plurality of curved edges.
4. The pull station housing of claim 1, wherein the alarm assembly further having: a front, a back, a top, a bottom, a left side, and a right side, wherein the right side and the left side have openings to emit light from an interior light source.
5. The pull station housing of claim 1, wherein the cover having an exterior surface and an interior surface, wherein the alarm assembly is secured to the interior surface of the cover by a plurality of nubs to securely hold the alarm assembly by an interference fit.
6. The pull station housing of claim 1, wherein the frame having: a front side and a back side; a groove along the perimeter of the back side of the frame to receive a sealing member to form a seal between the frame and the baseplate; and a tongue along the perimeter of the front side of the frame

to engage the cover groove.

7. The pull station housing of claim 1, wherein the cover further having: a front side and a back side; a groove along the perimeter of the back side of the cover to receive a cover sealing ring to form a seal between the cover and the frame.

8. The pull station housing of claim 1, wherein the frame further having: a top curved cut-out; a bottom curved cut-out; the top and bottom curved cut-outs having a predetermined radius and grooves.

9. The pull station housing of claim 8, further having: a conduit insert sized and shaped to engage the top and bottom curved cut-outs, wherein the conduit insert conforms to the shape of the curved cut-outs for a conformal fit.

10. The pull station housing of claim 9, wherein the conduit insert is made of foam like material.

11. The pull station housing of claim 1, further having: a subplate configured within the frame and separating the alarm pull station from the baseplate, the subplate comprising a subplate cutout revealing at least a portion of the alarm pull station to the baseplate.

12. The pull station housing of claim 11, further having a plurality of subplate offsetters separating the subplate and the baseplate.

13. The pull station of claim 1, wherein the alarm assembly further having a delay mechanism that, when activated, causes the alarm assembly to cease emitting noise for a pre-programed duration.

14. A fire alarm pull station housing comprising: a baseplate frame removably attached to a surface forming a perimeter around an alarm pull station; a cover hingedly attached to the baseplate frame; a handle secured to the cover; a magnet engagement means; an alarm assembly, wherein when a pull force is exerted upon the handle sufficient to overcome the resistance of the magnetic engagement means, the cover is displaced laterally away from the frame in an open position triggering the alarm assembly; and the magnetic engagement means securing the cover to the frame when no pull force is exerted.

15. The pull station housing of claim 14, wherein a sealing member is interposed between the cover and the baseplate frame, and wherein the magnetic engagement means urges the cover toward the frame, compressing the sealing member, thereby forming a substantially airtight and dust-resistant seal.

16. The pull station housing of claim 14, wherein the cover having: a plurality of curved edges; and an exterior surface and an interior surface, wherein the alarm assembly is secured to the interior surface of the cover by a plurality of nubs to securely hold the alarm assembly by an interference fit.

17. The pull station house of claim 14, wherein the alarm assembly further having: a front, a back, a top, a bottom, a left side, and a right side, wherein the right side and the left side have openings to emit light from an interior light source.

18. A fire alarm pull station housing comprising: a baseplate frame removably attached to a wall forming a perimeter around an alarm pull station; a cover hingedly attached to the baseplate frame, wherein the cover has an exterior surface and an interior surface; a handle secured to the cover; a magnetic engagement means; an alarm assembly, wherein when a pull force is exerted upon the handle sufficient to overcome the resistance of the magnetic engagement means, the cover is displaced laterally away from the subplate frame in an open position triggering the alarm assembly; the magnetic engagement means securing the cover to the frame when no pull force is exerted; and the alarm assembly is secured to the interior surface of the cover by a plurality of nubs by an interference fit.

19. The pull station housing of claim 18, wherein the alarm assembly further having: a front, a back, a top, a bottom, a left side, and a right side, wherein the right side and the left side have openings to emit light from an interior light source.

20. The pull station of claim 18, wherein the alarm assembly further having a delay mechanism

that, when activated, causes the alarm assembly to cease emitting the noise for a pre-programed duration.
